Bengali Fake News and Hate Speech

Evidence-Aware Classification and Evaluation Workflow

INPUT CORPORA

Fake news: 94,883 articles | BD-SHS: 50,281 comments
Original binary source labels

PREPROCESSING AND FROZEN SPLITS

TF-IDF: aggressive cleaning | Transformers: light normalization
FN: 75,905 / 9,489 / 9,489 | HS: 40,224 / 5,028 / 5,029
Post-hoc overlap flags: FN 48; HS 32 test rows

CLASSICAL BASELINES

TF-IDF: up to 5,000 terms
Logistic regression and Linear SVM
Naive Bayes and Random forest
Separate baseline comparisons

TRANSFORMER FINE-TUNING

BanglaBERT and multilingual BERT
2 datasets x 2 encoders x 3 seeds
42, 52, 62 = 12 experimental runs
Maximum length: FN 256 / HS 128

FROZEN TEST PREDICTIONS

Original-label test sets: FN 9,489 | HS 5,029

STATISTICAL EVALUATION

Multi-seed accuracy and macro-F1
Paired McNemar and Holm correction
Full-test length stratification
Source-held-out CPU diagnostic

EXPLAINABILITY

Seed-42: 20 FN + 20 HS cases
SHAP, LIME, Integrated Gradients
Random masking and sufficiency
Auxiliary Option B: 12 articles
(new sliding-window aggregator)

HUMAN-LABEL AUDIT

360 FN + 240 HS samples
Two independent authors
Third Hate Speech reviewer
Restricted sensitivity analysis
FN 259 / HS 192 cases

EVIDENCE-BOUND REPORTING

Original-label performance + restricted human-label sensitivity
Explicit truncation, overlap and auxiliary-system limitations